
CG000: BRANCH-RELATIVE ORIGIN RESOLUTION

COGITANT WHITEPAPER

Justin K. Lietz 
Neuroca, Inc.
justin@neuroca.ai

March 13, 2026

ABSTRACT

This Cogitant records a serious exploratory line of thought arising from the H000 / CF000 root pivot. It considers the possibility that the primitive bifurcation currently identified as the earliest serious root for the present derivational branch may still be only the earliest *accessible* origin rather than the absolute earliest origin of all possible reality. Under that possibility, major domain interfaces may reveal antecedent structure at higher resolution, so that what appears primitive within one branch later becomes visible as a coarse projection of a deeper origin distinction. The document does not advance this as a hypothesis, conjecture, or formal claim. It preserves the thought in a structured whitepaper form so it can be pursued later without contaminating the sharper H000 root program.

Keywords branch-relative origin · cogitant · hierarchy · domain interfaces · H000 · CF000 · VDM

1 Document status and objective

Classification	Cogitant / exploratory whitepaper
Status	ACTIVE; structured exploratory document, not a hypothesis, conjecture, or formal claim
Scope	This Cogitant explores intuition and considers possibilities without yet settling into a hypothesis, conjecture, or formal claim.
One-line objective	Consider whether the primitive bifurcation currently identified in H000 / CF000 is the earliest accessible origin for the present derivational branch rather than the absolute earliest origin of all possible reality, and whether major domain interfaces progressively reveal earlier antecedent structure at higher resolution.
Canon placement	CG000 sits adjacent to the H000 / CF000 root program as a preserved exploratory direction. It does not replace H000; it records what may lie beyond it if the present root is branch-local rather than absolute.

2 Why this thought matters

H000 currently functions as the earliest serious root candidate available to the present derivation program. That is already a major step forward. But the fact that H000 may be the earliest accessible origin for this branch does not necessarily imply that it is the absolute bottom of all possible origin structure.

This thought matters because it addresses a live pressure that appears whenever a derivation program successfully reaches a new root: the new root may still feel too late, but the next deeper layer may be inaccessible from the current branch because the system lacks sufficient resolution, observability, or available degrees of freedom. If that is true, then “origin” may be branch-relative rather than absolute, and the project may need a principled way to speak about deeper unresolved antecedents without corrupting the current root work.

This also matters because it may connect the primitive-origin program to the hierarchy program in a new way. If branch growth outward and origin refinement backward are structurally linked, then the hierarchy logic currently aimed at interface formation, truncation, and branch depth may have a deeper orthogonal analogue.

3 Motivating intuition

The motivating intuition is simple: after pushing the origin derivation back as far as currently possible, the thought naturally arises that the limiting factor may no longer be logic alone, but visibility. If a deeper origin exists, it may not be directly reachable from the current branch because the current domain does not have enough resolution or enough degrees of freedom to resolve it.

Under that intuition, the present primitive bifurcation might be the earliest visible distinction from inside the current branch, while a deeper antecedent distinction would only become visible in a richer codimension- $N + 1$ regime. If so, the emergence of a new domain or interface might act like a mirror that reveals a sharper version of what currently appears only as a coarse primitive split.

This suggests a strange inversion: the farther one advances away from a perceived origin point through emergent structure, the more that origin point may regress into an earlier antecedent origin. What appears primitive at one level may later reveal itself as only a low-resolution face of a deeper branch structure.

4 Exploratory chain of thought

The thought begins from a pressure already present in the current work. H000 / CF000 were forced into existence because previous candidate roots kept starting too late. QGT was too rich. Carrier structure was too rich. Geometry was too rich. Even elegant and powerful mathematical objects still looked like downstream products rather than true beginnings.

H000 answers that pressure by moving to a primitive bifurcation between sterile complementary poles and deriving distinguishability and hierarchy before geometry. But once that move is made, a new pressure appears: perhaps this “primitive” origin is only primitive relative to the present branch.

That immediately raises a new possibility. The earliest visible distinction in one branch may not be the earliest possible distinction in all branches. Instead, branch-relative origin may depend on what kinds of interfaces, codimensions, or observational freedoms are available from within the domain. A deeper origin might therefore remain invisible until a later domain interface gives the system enough structural resolution to expose a sharper antecedent split.

From there the intuition becomes more geometric and hierarchical. Each major interface transition may re-express the origin at a higher resolution. The current primitive bifurcation could therefore be a coarse mirror of a deeper unresolved distinction. If that continues, the structure of origin itself may be recursive: not a single final root, but a hierarchy of branch-relative roots.

That in turn suggests an inverse analogue to the current hierarchy conjecture. In the usual direction, hierarchy branches forward into structure, interfaces, and late physical regimes. In the inverse direction, origin may regress backward asymptotically into deeper antecedent structure as one moves into richer branches or more resolved domain interfaces. If there is a truncation rule on forward hierarchy depth under finite admissibility, perhaps there is also a truncation or asymptotic resolution rule on backward origin refinement.

This does not yet imply an infinite regress in a useless sense. One possibility is that each branch has a finite accessible origin depth, and that beyond some point the difference between deeper antecedents becomes too small relative to the available resolution of that branch. Another possibility is that the system is recursively self-similar, so that what looks like a completed hierarchy within one branch becomes only one lobe of a larger branch structure, almost like a Mandelbrot-style recurrence across domain types and degrees of freedom.

If any of this is right, then H000 may still remain correct as the right root for the present branch, while also not being the ultimate bottom. That would be an important distinction: a root can be both locally correct and globally incomplete.

5 Relation to current canon

This Cogitant sits *beneath and beyond* the current H000 / CF000 root program.

It sits beneath it in the sense that it questions whether H000 is absolute or branch-relative. It sits beyond it in the sense that H000 itself should remain the current operational root hypothesis unless and until a stronger earlier derivation becomes accessible.

- **H000 / CF000.** This Cogitant does not replace H000. It treats H000 as the current earliest serious branch-root and asks whether that root may itself be a coarse accessible projection of a deeper antecedent origin.
- **CF00.** If branch-relative origin resolution is real, then CF00 becomes even more clearly a downstream branch formalism that begins only once a carrier-domain regime has already stabilized.
- **A8 / hierarchy work.** This Cogitant is strongly related to the existing hierarchy program. It raises the possibility that forward hierarchy depth and backward origin refinement obey related structural laws along orthogonal directions.
- **Runtime pressure.** The same runtime pressure that forced the move past QGT may eventually force a later move past H000 if a new branch or interface exposes hidden antecedent structure.

This Cogitant therefore does not conflict with current canon so much as it introduces a possible meta-architectural extension: branch-local roots may themselves live inside larger recursive origin hierarchies.

6 Possible hypothesis directions

Potential future hypothesis directions include:

- **Branch-relative origin hypothesis:** the earliest accessible origin for a derivational branch is not necessarily the absolute earliest origin of all possible reality.
- **Interface-resolution hypothesis:** major domain interfaces increase the resolution with which antecedent origin structure can become visible.
- **Orthogonal hierarchy hypothesis:** forward hierarchy branching and backward origin refinement obey related truncation or scaling principles along different structural axes.
- **Asymptotic origin-resolution hypothesis:** deeper origin layers become progressively less distinguishable from within a given branch, producing an asymptotic limit on backward resolution.
- **Recursive branch hypothesis:** what appears as a completed hierarchy within one branch may itself be only one branch inside a larger hierarchy of origin-and-domain formation.

These are not being asserted here. They are candidate future directions if the current intuition sharpens.

7 Risks and failure modes

Risk 1 — endlessly receding earliness. This Cogitant could become too unconstrained and slide into an endlessly receding “there is always something earlier” posture. That would make it difficult to use scientifically and would weaken the present root program instead of strengthening it.

Risk 2 — metaphor remaining metaphor. The language of “mirror,” “higher resolution,” and “deeper origin” could remain evocative rather than becoming structurally meaningful. If the idea cannot produce disciplined consequences for interface behavior, hierarchy, or branch-relative observability, then it may remain only an image.

Risk 3 — contamination of H000. If this line of thought is folded prematurely into the current root hypothesis, it could blur the sharpness of H000 and make the present derivation harder to audit. H000 should remain the current working root unless and until a cleaner earlier burden is actually formalized.

Risk 4 — false self-similarity. Reality may not recurse in the elegant way this Cogitant suggests. The resemblance to Mandelbrot-like branching may reflect pattern hunger rather than lawful structure.

8 Status

Current status. ACTIVE.

Current use. This Cogitant exists to preserve and formalize a strong exploratory intuition that arose directly from the success of the H000 pivot. It is not a replacement for H000, but a structured note on what may lie beyond it if the current branch is not absolute.

Promotion condition. This Cogitant should only promote into a hypothesis if it can generate disciplined future burdens, such as explicit interface-level predictions, branch-relative observability criteria, or a clean orthogonal relation to the hierarchy truncation program.

Notes

This Cogitant may eventually motivate a formal bridge paper connecting origin-resolution limits, hierarchy-depth truncation, codimension growth, and domain-interface transitions.

It is also a useful place to record the intuition that major interfaces such as false-vacuum transitions, horizon-like boundaries, or still deeper carrier-support transitions may act as windows onto antecedent branch structure.

Version history. v0.1 (2026-03-13) — created from the initial CG000 markdown artifact and formatted into a whitepaper preserving the H000 style while applying the looser CG standard.